

# Differentiated Carbon Prices or International Transfers?

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Countries commit to a **carbon price floor, increasing with their income level**, and  
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However **efficiency requires a uniform price**, and equity concerns can be settled by **transfers**.

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We compute the emissions rights equivalent to prominent differentiated prices proposals.

# Differentiated prices *vs.* a uniform price

## Why differentiate? Four imperfections

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⇒ **Inconsistent with survey evidence** (Fabre et al., 2025; Fabre, 2026).

None of the four decisively discard a uniform price: 1. uncalibrated, 2. irrelevant, 3. compatible, 4. unproven.

# Standard Theory

## The static case

Country  $i$  produces  $y_i$ , consumes  $c_i$ , and at carbon price  $p_i$  faces abatement cost  $a_i$ :

$$c_i = y_i - a_i + t_i, \quad t_i = \text{international transfer.}$$

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$p^*$ : uniform price attaining  $E^*$ , i.e.  **$p^*$  s.t.  $E^* = \sum_j e_j(p^*) n_j$ .**

Shorthands  $e_i^* := e_i(p^*)$ ,  $a_i^* := a_i(e_i^*)$ .

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$u_i$  can be expressed as function of  $s$ : we have  $u_i(0, p_i) = c_i^a = y_i - a_i$ , and  $u_i(t, p^*) = c_i^* + t = y_i - a_i^* + t$ .

## Step 1: the equivalent transfer

$V_i(p_i)$ : the lump sum transfer which makes  $i$  as well off with the global price as in autarky:

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Optimising country sets  $\frac{da_i}{de}(e_i^*) = -p^*$ : Therefore, a first-order Taylor expansion around  $e_i^*$  gives:

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**Equivalent transfer**  $g_i^s$  of an arbitrary situation  $s = (t_i, p_i)$ , relative to the benchmark:

$$g_i^s = \underbrace{t_i}_{\text{cash received}} + \underbrace{V_i(p_i)}_{\text{gain from facing } p_i \neq p^*}$$

## Step 2: the correspondence between prices and rights

Cap-and-trade at  $p^*$ , each person in  $i$  endowed with rights  $r_i$ . Net monetary transfer to  $i$ :

$$t_i^* = (r_i - e_i^*) p^* \quad \implies \quad g_i^* = t_i^*.$$

Autarky, from Step 1:  $g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*$ . Setting  $g_i^* = g_i^a$ :

$$\begin{aligned} & \boxed{(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*} \\ \iff & \quad r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^* \quad \iff \quad p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*. \end{aligned}$$

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**The key formula for policy design:** rights  $r_i$  under uniform price equivalent to autarky price  $p_i$

Low autarky price ( $p_i < p^*$ )  $\Rightarrow$  more rights than emissions  $r_i > e_i^*$  to compensate the extra abatement cost.

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If the two can always be made equivalent, *is there any reason to prefer one?* Yes.

### Step 3: the uniform price weakly dominates, for *every* country

Give country  $i$  rights equal to its *autarky* emissions,  $r_i = e_i$ . Net transfer  $\tilde{t}_i = (e_i - e_i^*)p^*$ .

Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = -(e_i - e_i^*)p^* \quad \implies \quad \tilde{t}_i \geq a_i^* - a_i = V_i(p_i) = g_i^a.$$

Hence, consumption increases *in every country* when trading is allowed and prices equalize:

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This is Diamond & Mirrlees (1971) production efficiency applied to climate: absent other distortions, direct transfers beat differentiated factor prices as a redistributive instrument.



## The dynamic case: the equivalence breaks

Add time  $t$ ; each variable becomes a trajectory.  $\beta_t$  = discount factor,  $R_i = \sum_t r_{it}$  =  $i$ 's carbon budget;  $G_i = \sum_t g_{it}\beta_t$  : total discounted equivalent gain.

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Welfare equivalence now requires equality of *entire discounted streams*,  $G_i^U = G_i^A$ :

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A **generous** allocation in a **low-price** period is **worth less** than the same allocation when prices are high.

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$$\mathcal{B} := \sum_t r_t \beta_t p_t^* \quad \rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t,$$

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Analogous to static formula, but  $\mathcal{B}$  still requires the full price path  $\Rightarrow$  **not tractable without a model.**

## Special case 2: Hotelling

Hotelling rule: Fixed aggregate carbon budget + perfect foresight + efficient capital markets  
 $\Rightarrow$  carbon price rises at the rate of return on capital. Discounting at that rate,  $\beta_t p_t^* = p_0$  is constant.  
 $\beta_t p_t^*$  then factors out of both sums, and the dynamic condition collapses to:

$$R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* \quad E_i^* = \sum_t e_{it}^*$$

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In the numerical analysis, we place ourselves in the Hotelling case.

# Numerical analysis

## **Taking the formula to a calibrated model**

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Grid search over  $(\rho_i, \pi_i)$  for eight major regions: **USA, Russia, EU27, China, Turkey, India, Nigeria, DRC**.

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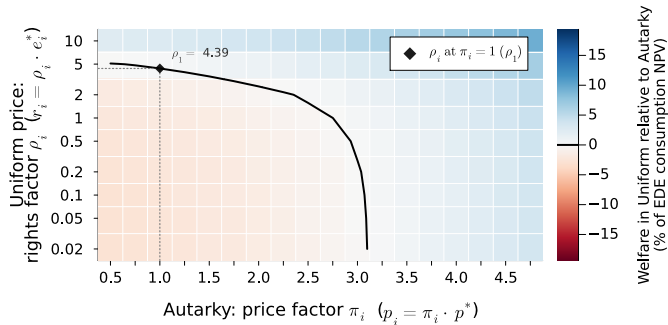
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RoW receives rights so that global emissions coincide with autarky, grandfathered on baseline emissions.

## Reading the heatmaps

USA welfare **gain of uniform over autarky**, depending on rights and price.

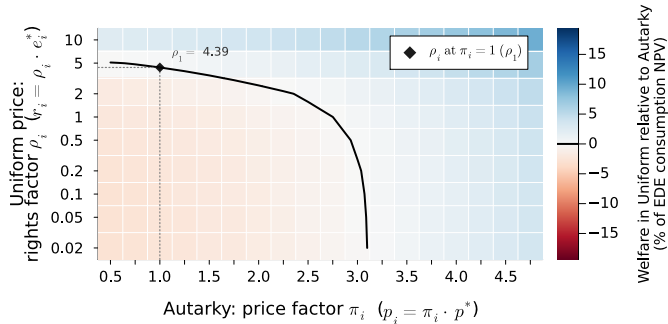


Grid showing the welfare **gain** or **loss** of *Uniform over Autarky*

**Blue:** Uniform price with transfers dominates; **Red:** Autarky dominates; line: indifference curve.

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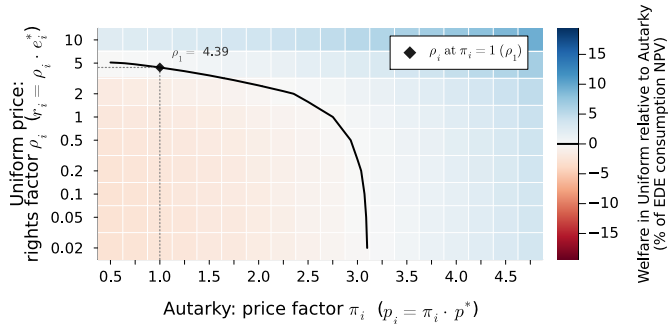
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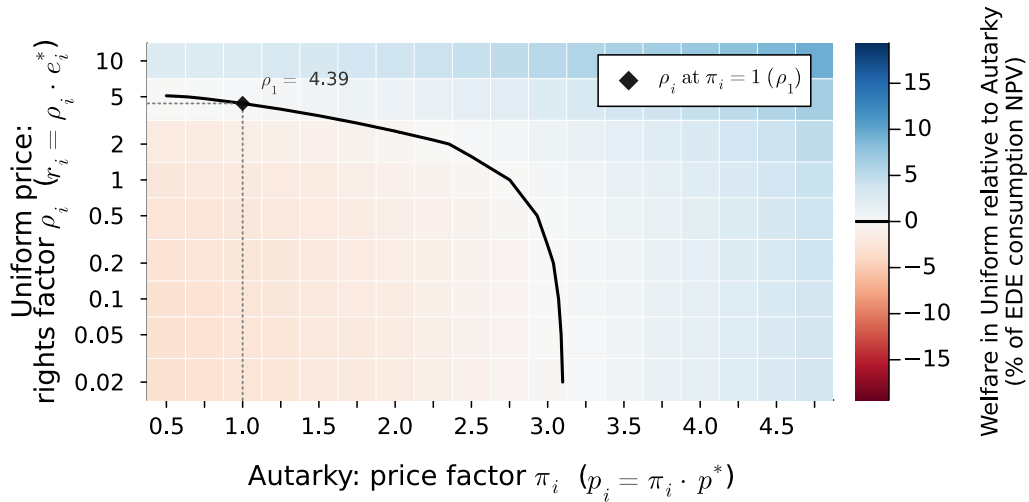
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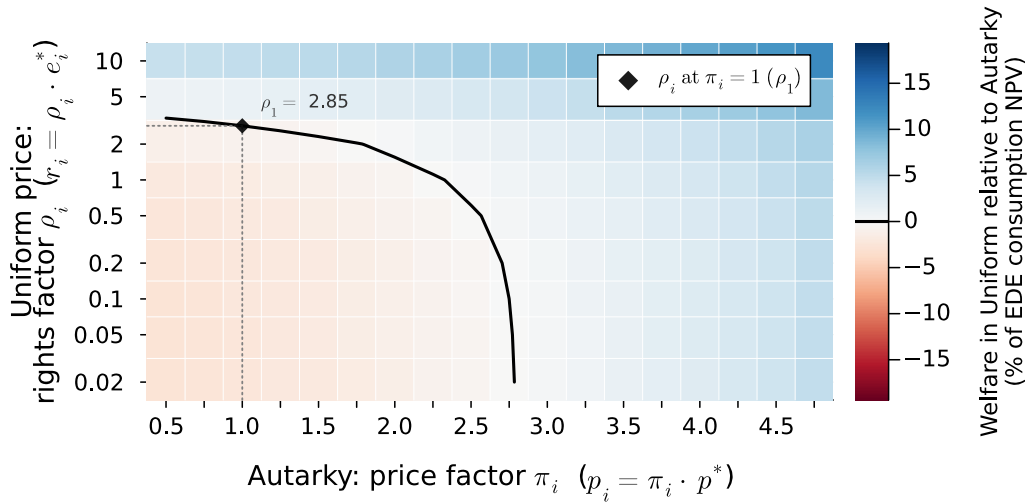
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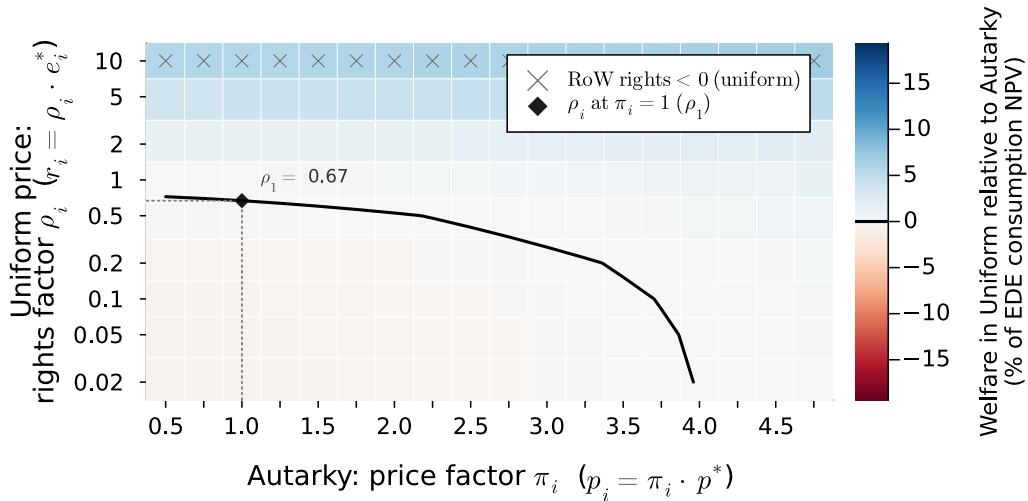
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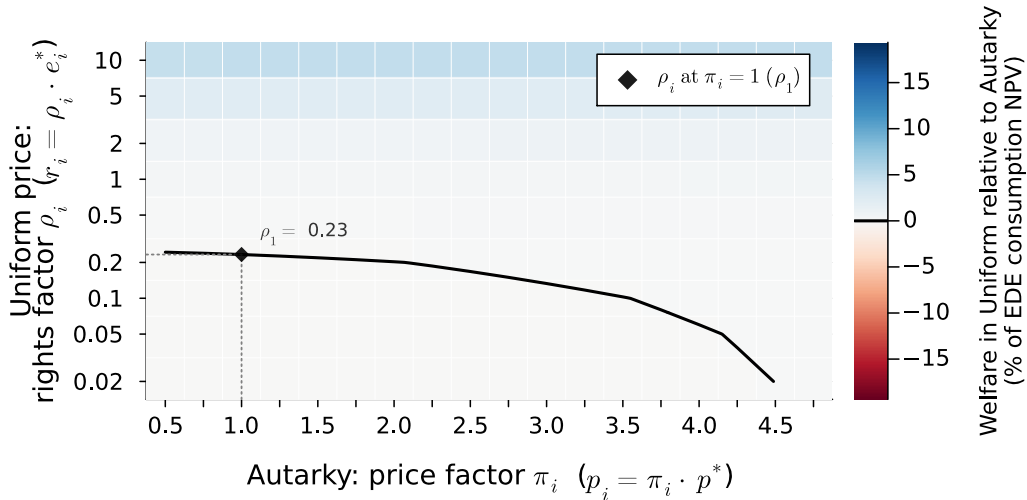
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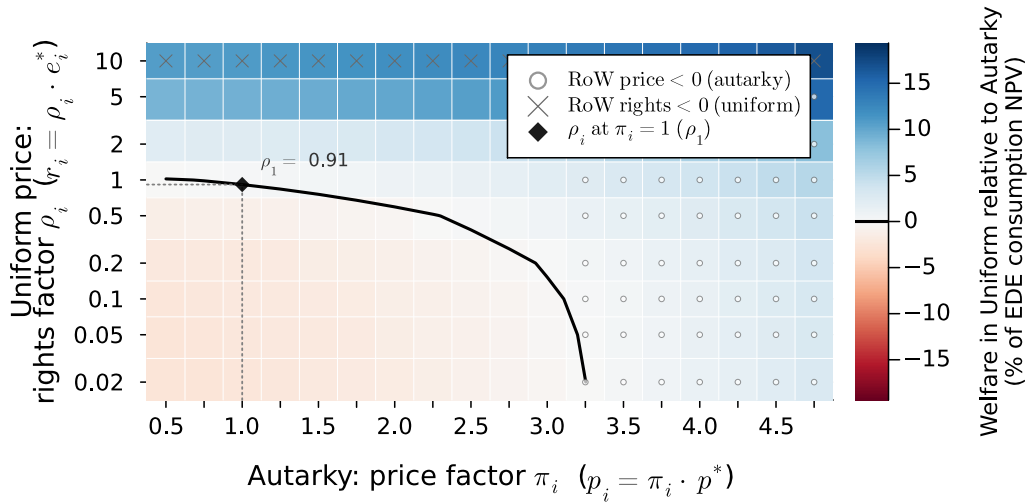
## China



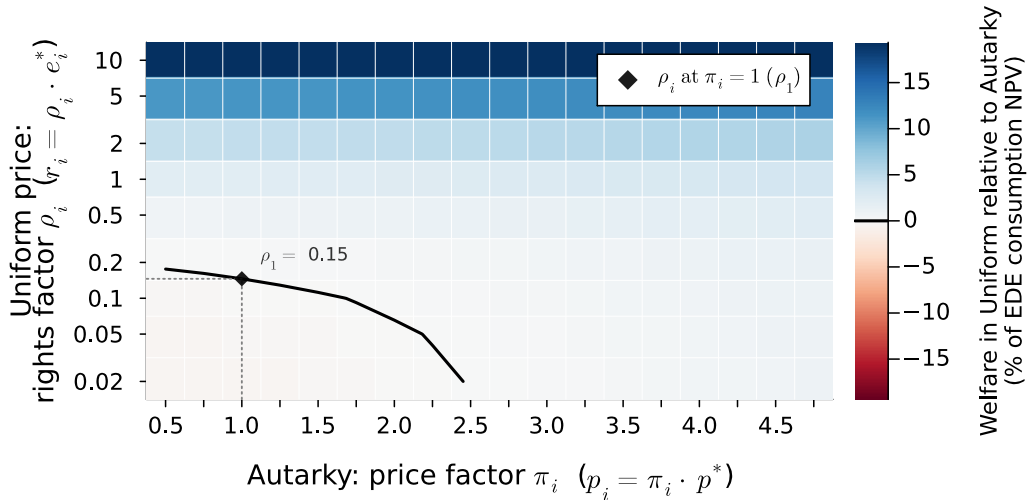
## European Union (EU27)



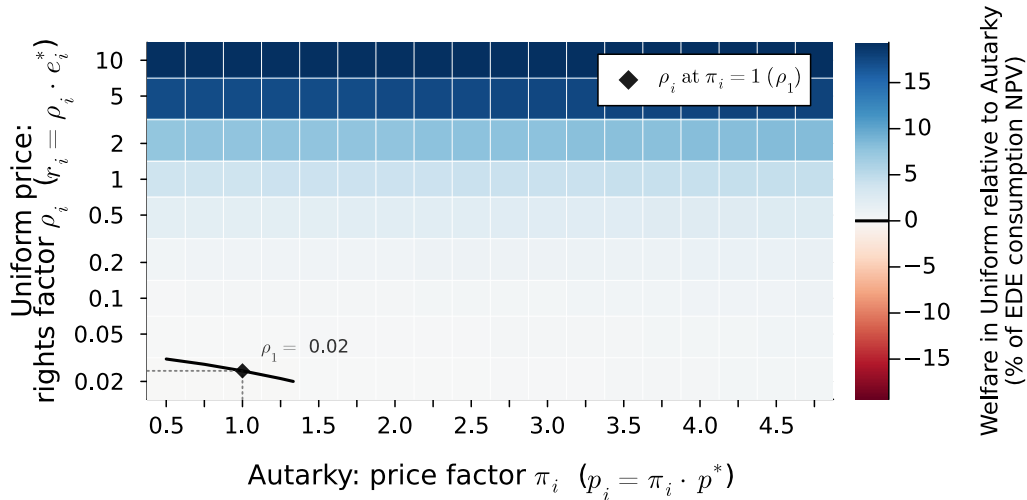
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## Democratic Republic of the Congo





# Assessment of recent proposals

## The proposals

IMF; Working Group on Climate Coalitions propose a coalition with price floors differentiated by income: \$75/tCO<sub>2</sub> high-income, \$50/t upper-middle, \$25/t lower-income (Parry et al., 2021; Wolfram et al., 2025).

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Banerjee, Duflo & Greenstone (2025) propose a grand bargain with \$10 LICs, \$30 lower-MICs, \$50 upper-MICs (no price asked of HICs, I assume \$75/t).

Pricing gives right to climate-damage compensation, funded by HICs through global taxes on wealth and profit.

## Equivalent rights for the proposals

| Country                                                                                 | Wolfram et al. |          | Banerjee et al. |          |
|-----------------------------------------------------------------------------------------|----------------|----------|-----------------|----------|
|                                                                                         | $p_i$          | $\rho_i$ | $p_i$           | $\rho_i$ |
| United States                                                                           | 0              | –        | 75              | 2.16     |
| Russia                                                                                  | 0              | –        | 75              | 1.36     |
| European Union                                                                          | 75             | 0.26     | 75              | 0.26     |
| Turkey                                                                                  | 0              | –        | 50              | 1.54     |
| Indonesia                                                                               | 50             | 0.89     | 50              | 1.09     |
| China                                                                                   | 50             | 0.84     | 50              | 0.88     |
| Brazil                                                                                  | 50             | 0.38     | 50              | 0.5      |
| India                                                                                   | 25             | 1.3      | 30              | 1.18     |
| Nigeria                                                                                 | 0              | –        | 30              | 0.24     |
| DR Congo                                                                                | 0              | –        | 10              | 0.06     |
| <i>Equivalent rights as solved: the allocation sets the cap</i>                         |                |          |                 |          |
| Reduced emissions in the coalition (%)                                                  |                | 9.9%     |                 | 9.1%     |
| World temp. 2100, change (°C)                                                           |                | –0.020   |                 | –0.037   |
| World welfare gain (%)                                                                  |                | 0.098%   |                 | 0.208%   |
| World consumption gain (%)                                                              |                | 0.037%   |                 | 0.050%   |
| <i>Rights raised to the coalition's emissions, shared to maximise the smallest gain</i> |                |          |                 |          |
| World welfare gain (%)                                                                  |                | 0.023%   |                 | 0.095%   |
| World consumption gain (%)                                                              |                | 0.021%   |                 | 0.056%   |

$p_i$ : the price the proposal asks of  $i$  in 2030 (\$/t), against a single price for all coalition members of \$49.2/t (Wolfram et al.) and \$55.8/t (Banerjee et al.); countries the proposal does not price stay outside the coalition ( $p_i = 0$ , no allocation). The proposals themselves reach 2.16°C (Wolfram et al.) and 1.80°C (Banerjee et al.) in 2100. In the first block the equivalent rights are used as they are solved, so the coalition emits less than under the proposal; in the second they are raised to the proposal's own emissions and shared so as to make the worst-off member as well off as possible



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**Thank you for your attention!**

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# Appendix

Two boundary conditions evaluated in *every* year of 2030–2100:

1. **Market-distorting subsidies** ( $\omega_i \times \pi_i > 1$ ): the target country claims more than 100% of the global price burden. To compensate, the model is forced to push  $p_{-i} < 0$  — economically, a **fossil-fuel subsidy for the RoW**.

2. **Negative emission rights** ( $\rho_i > 1 / pop_i$ ): granting  $i$  its multiplier uses more than the whole global budget, forcing the RoW into **negative rights**. Binds for large populations: countries above 10% of world population (China, India) trigger it for any  $\rho_i > 10$ .

---

**RoW price identity**, the starting point of the root-finding:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}, \quad \omega_i = e_i / E^* \quad \implies \quad p_{-i} = p^* (E^* - e_i \pi_i) / e_{-i}.$$

This linear approximation assumes emission-weighted price averaging delivers the same global abatement — false with non-linear MACs, hence the **Newton–Raphson** correction to  $< 0.1\%$  relative error on the cap.

## Autarky scenario, United States ( $\pi = 2$ ) [Go back](#)

| Variable                      | 2030 | 2040  | 2050  | 2060  | 2070  | 2080  | 2090  | 2100  |
|-------------------------------|------|-------|-------|-------|-------|-------|-------|-------|
| $p^*$ (USD/tCO <sub>2</sub> ) | 42.6 | 112.9 | 205.9 | 285.7 | 401.6 | 480.9 | 476.2 | 471.4 |
| $p_i$ (USD/tCO <sub>2</sub> ) | 85.2 | 225.8 | 411.7 | 571.4 | 803.3 | 961.9 | 952.3 | 942.8 |
| $E_i$ (GtCO <sub>2</sub> )    | 4.31 | 2.58  | 0.63  | -0.53 | -1.75 | -2.22 | -1.86 | -1.52 |

Net-negative US emissions from the 2060s: abatement rates  $\mu_i > 1$ , i.e. [CDR deployment](#), with the MAC function extrapolated beyond the 100% boundary.

Standard IAMs constrain  $\mu_i \in [0, 1]$ ; enforcing that bound crashes the grid search under extreme policy parameters.

## The equivalent price, and why we do not use it [▶ Go back](#)

One could instead define an **equivalent price**  $p_i^*$  such that  $(0, p_i^*)$  is welfare-equivalent to  $(t_i, p_i)$ :

$$t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^* \quad \implies \quad p_i^* \approx p_i - \frac{t_i}{e_i^*\varepsilon^*}.$$

**Problem:**  $p_i^*$  can be *negative* when the transfer is large, which has no natural interpretation as a carbon price.

The equivalent transfer  $g_i^s$  does not share this limitation  $\implies$  we use it as the primary welfare metric.

**Last slide (needed)**